

When Benchmark Inferences Do Not Compose: Projectibility in AI Evaluation

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DECLARATIONS

Availability of data and material. The empirical companion, including pinned source versions, data hashes, and estimator code, is publicly available at <https://github.com/BrettRey/benchmark-inference-composition>. The reanalysed trial data are the public release accompanying Zhang, Koyejo, and Yang (2026). No new human-subject, proprietary, or restricted data were generated. The constructed results in Section 4 are stipulated for illustration, not observations of any deployed system, and the design that would produce them is stated in full.

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Ethics approval. Not applicable. The study involves no human participants and no animals.

Consent to participate and consent for publication. Not applicable.

Use of large language models. The large language models ChatGPT 5; Claude Sonnet 4.5 and Opus 4.8; Gemini 3.1 Pro served as drafting and editing aids throughout the preparation of this paper. Their use went beyond AI-assisted copy editing and is declared here in full. The author is responsible for all theoretical claims, arguments, errors, and interpretive choices, and affirms that the final text reflects his own work. No language model is listed as an author, and none satisfies the journal's authorship criteria.

Prior dissemination. A preprint of this work is posted at [arXiv:2607.26159](https://arxiv.org/abs/2607.26159). The manuscript is not under consideration at any other journal, and no part of it has been published elsewhere.